

Subtraction peri-ictal SPECT is predictive of extratemporal epilepsy surgery outcome

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Article abstract—*Objectives:* To determine whether localization of extratemporal epilepsy with subtraction ictal SPECT coregistered with MRI (SISCOM) is predictive of outcome after resective epilepsy surgery, whether SISCOM images provide prognostically important information compared with standard tests, and whether blood flow change on SISCOM images is useful in determining site and extent of excision required. *Background:* The value of SISCOM in predicting surgical outcome for extratemporal epilepsy is unknown, especially if MRI findings are nonlocalizing. *Methods:* SISCOM images in 36 consecutive patients were classified by blinded reviewers as “localizing and concordant with site of surgery,” “localizing but nonconcordant with site of surgery,” or “nonlocalizing.” SISCOM images were coregistered with postoperative MRI, and reviewers visually determined whether cerebral cortex underlying the SISCOM focus had been completely resected, partially resected, or not resected. *Results:* Twenty-four patients (66.7%) had localizing SISCOM, including 13 (76.5%) of those without a focal MRI lesion. Eleven of 19 patients (57.9%) with localizing SISCOM concordant with the surgical site, compared with 3 of 17 (17.6%) with nonlocalizing or nonconcordant SISCOM, had an excellent outcome ($p < 0.05$). With logistic regression analysis, SISCOM findings were predictive of postsurgical outcome, independently of MRI or scalp ictal EEG findings ($p < 0.05$). The extent of resection of the cortical region of the SISCOM focus was significantly associated with the rate of excellent outcome (100% with complete resection, 60% with partial resection, and 20% with nonresection, $p < 0.05$). *Conclusion:* SISCOM images may be useful in guiding the location and extent of resection in extratemporal epilepsy surgery.

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Peri-ictal blood flow SPECT is increasingly being established as an accurate and reliable test in the preoperative localization of intractable partial epilepsy.^{1–10} Furthermore, our group and others have found that the clinical utility of peri-ictal SPECT can be enhanced by using computer-aided techniques that produce peri-ictal/interictal subtraction images (difference images) coregistered with MRI (SISCOM).^{11–16} Previous studies have demonstrated that the findings on SISCOM analysis of peri-ictal and interictal scans are predictive of the outcome of epilepsy surgery in a mixed group of patients with temporal or extratemporal partial epilepsy.^{11,12} A majority of patients in these earlier studies had temporal lobe epilepsy (TLE) surgery. The temporal lobe surgical focus usually can be well localized with standard techniques, particularly with ictal video-EEG recording and seizure-protocol MRI, and the outcome after resection generally has been good.^{17,18}

Successful epilepsy surgery is more elusive in extratemporal lobe epilepsy (ETLE) than in TLE. The surgical epileptogenic zone is often more difficult to localize precisely, particularly in the absence of a

potentially resectable MRI-detected lesion. Scalp EEG, interictal SPECT, and fluoro-18-deoxyglucose (FDG)-PET are often less localizing in ETLE than in TLE. Consequently, outcome after ETLE surgery is significantly poorer. The rate of freedom from seizure after extratemporal surgery (excluding lesionectomy) is 30% to 50%,^{18–21} whereas it is $\geq 80\%$ after TLE surgery.^{22–26} Although a standard resection is often performed in most cases of TLE surgery (i.e., usually an anterior temporal lobectomy and amygdalohippocampectomy), the extent of resection required in most cases of ETLE is not well established. A large resection increases the chance of rendering the patient seizure-free, but it also increases the potential for postoperative morbidity.²⁷

Better methods are required to identify more accurately the site and the extent of the surgical epileptogenic zone in ETLE. Peri-ictal SPECT and SISCOM analysis have great potential for improving surgical localization in this group of patients with complex and intractable epilepsy, particularly if the methods are used with stereotactic image-guided surgical techniques. However, very few studies have

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specifically assessed the usefulness of peri-ictal SPECT in ETLE, and most of these involved a small number of patients.⁷⁻¹⁰ Furthermore, no previous study has specifically validated the use of computer-aided subtraction SPECT techniques in ETLE.

We investigated 1) whether the localization of extratemporal epilepsy with the SISCOM technique is predictive of outcome after resective epilepsy surgery; 2) whether SISCOM images provide prognostically important information in addition to that provided by standard tests; and 3) whether the area of blood flow change on SISCOM images is useful in determining the site and the extent of the excision required for successful postoperative seizure control.

Methods. *Patients, demographics, and surgical details.* The study group consisted of 36 consecutive patients who had peri-ictal and interictal SPECT studies and extratemporal resective epilepsy surgery performed at Mayo Clinic Rochester between June 1993 and June 1997. All patients had at least 12 months of postoperative follow-up. The study was approved by the Institutional Review Board of Mayo Foundation. During the study period, 257 patients had ictal SPECT studies performed as part of the evaluation for intractable partial epilepsy. Of these patients, 111 subsequently had epilepsy surgery (65 had temporal lobe resection and 46 had extratemporal lobe resection exclusively or partially). Of the extratemporal lobe group, five were excluded from the study because they did not have interictal SPECT performed, and five others were excluded because they did not have at least 12 months of postoperative follow-up.

The study cohort included 23 male and 13 female patients, with a median age of 25 years at the time of surgery (range, 1.5 to 56 years). The site of cortical excision was frontal in 19 patients, frontoparietal in 4, occipital in 4, frontotemporal in 2, parietal in 2, parietotemporal in 2, parieto-occipital in 2, and occipitotemporal in 1. Fifteen patients (41.7%) had maximal frontal resection, 6 (16.7%) had limited corticectomy, 14 (38.9%) had lesionectomy, and 1 (2.8%) had modified hemispherectomy. Maximal frontal resection in the nondominant hemisphere included the three horizontal frontal gyri over the lateral convexity and the cingulate gyrus around the corpus callosum. The posterior orbital cortex and the subcallosal gyrus were spared. In the dominant hemisphere, the resection was carried down to the middle of either the middle frontal or inferior frontal gyrus, depending on the location of the vascular structures in the area.²⁸

SPECT injection and imaging methods. Peri-ictal injections of radioisotope were performed by specially trained EEG technicians during typical seizures while the patient was undergoing inpatient video-EEG monitoring. The radioisotope ^{99m}Tc-HMPAO was injected in 14 patients (38.9%) and ^{99m}Tc-bicisate in the other 22 (61%). An interictal study was performed with the same radioisotope after the patient had been seizure-free for at least 24 hours, with standard ambient room lighting and the patient's eyes open and ears unplugged. The dose of the injected radioisotope for both the peri-ictal and interictal studies was approximately 20 mCi. We demonstrated previously that the magnitude of the peri-ictal increase in

focal cortical uptake was not significantly different between the two radioisotopes.²⁹

The SPECT images were acquired with the same scanner within 2 to 3 hours after the injection of radioisotope, using an identical protocol for both peri-ictal and interictal studies. A dual-headed gamma camera system (Helix system, Elscint Inc., Haifa, Israel) equipped with ultrahigh-resolution fan beam collimators was used. The data were acquired in a 128 × 128-byte matrix over 360 degrees, with 120 views obtained at 3-degree intervals using a circular orbit. For all studies, the gamma camera made five 360-degree orbits, with 3 seconds/view. The energy setting was 140 keV, with a 15% to 20% window. Transaxial images were reconstructed using a Metz filter (power = 3, full width at half maximum [FWHM] = 6 mm) rebinned into a 64 × 64 matrix with a 2× zoom. Attenuation correction (μ = 0.12 cm) was applied, and a standard series of contiguous images was created in the transaxial, coronal, sagittal, and transtemporal planes. The reconstructed system resolution was approximately 7 mm FWHM and was composed of cubic voxels with dimensions of 3.6 or 4.4 mm (depending on image size).

The time of the seizure and the time of the injection of the radioisotope were identified in the same manner as in our earlier studies,^{11,30} with the onset of a seizure defined as the time of earliest indication of a warning (verbalized or pushing the call button) or abnormal movements, behavior, or impaired awareness. The end of a seizure was defined as the time when the ictal movements or behavior ceased. When the start and the end of the seizure could not be established confidently from clinical features, ictal EEG was reviewed for the beginning and the end of the rhythmic seizure discharge. The time of the injection was defined as the time when the plunger of the syringe containing the radioisotope was fully depressed. The SPECT studies were subdivided into those in which the radioisotope was injected before the end of the seizure (ictal) and those in which the agent was injected after the seizure terminated (postictal).

SISCOM methods. Hyperperfusion SISCOM images were created as described previously.^{31,32} For patients who had postictal SPECT studies, the hypoperfusion images were created by multiplying the pixel values of the subtraction SPECT by -1 before thresholding, so that the highest pixel values in these images were those with the greatest decrease in intensity from the interictal study to the ictal study (i.e., maximal relative hypoperfusion).¹¹ Hyperperfusion images were thresholded to one SD level and hypoperfusion images were thresholded to both one and two SD levels. After thresholding, the images were coregistered with the patient's preoperative MRI for anatomic localization of the regions of blood flow change.¹¹

All patients in the study had a seizure protocol MRI performed 3 to 6 months after surgery. By using the surface matching technique for the SPECT-to-MRI coregistrations,³³ this image was segmented and then matched and transformed into the same three-dimensional space as the preoperative MRI. The SISCOM images were then coregistered with this transformed postoperative MRI, so that the relationship of the sites of uptake change on SPECT could be compared with the surgical defect.

Blinded SISCOM review for cortical localization. The SISCOM images were independently reviewed for regional

cortical localization by two primary reviewers (B.P.M., E.L.S.) who were blinded to the clinical data, results of other tests, and surgical outcome. The reviewers were required to localize the images to one of the 18 possible cerebral regions (i.e., either right or left frontal, frontotemporal, temporal, temporoparietal, parietal, occipitoparietal, occipitotemporal, or occipital) or to classify the images as “nonlocalizing.” The final determination was based on the agreement of the reviewers on the localization or nonlocalization of the images. If the two reviewers disagreed, the image was reviewed by a third blinded reviewer (M.F.H.). Agreement of the third reviewer with one of the primary reviewers served as the final determination. If the third reviewer failed to agree with either primary reviewer, the study was considered nonlocalizing.

Because this study formed part of a larger review of all epilepsy surgery patients who had peri-ictal SPECT studies, the reviewers did not exclude from consideration the option of temporal lobe localization. The reviewers were told the duration of the seizure and the injection timing. If the injection was ictal, only the two-SD hyperperfusion images were reviewed.¹² If the injection was postictal, both the one- and two-SD hypoperfusion images as well as the two-SD hyperperfusion images were reviewed. We showed previously that postictal blood flow changes are often diffuse and of a lower absolute intensity than ictal changes, and that the one- and the two-SD images yielded complementary information that was valuable in interpreting postictal SISCOM.¹¹

Comparison with the EEG and MRI localization. Ictal scalp EEG was recorded for all patients, using the 32-channel system with the electrodes arranged according to a modified 10–20 system, which included subtemporal electrodes. Of the 36 patients, 21 (58.3%) also had video-EEG monitoring with intracranial electrodes (12 had subdural strip/grid electrodes, 2 had bitemporal depth electrodes, and 7 had both bitemporal depths and subdural strips). Localizations by scalp EEG, intracranial EEG, and MRI were determined by retrospective review of the reports of the procedures. The localization rate by blinded SISCOM evaluation was compared with that by scalp EEG, intracranial EEG, and MRI. The agreement between SISCOM evaluation and these more established tests was also assessed.

Correlation of SISCOM localization with postsurgical outcome. Postsurgical outcome with respect to seizures was determined by review of the patients’ medical records. Outcome classification was based on the patients’ state of seizure control at latest follow-up. As in our earlier studies, outcome was classified according to a 12-point seizure frequency score (table 1).^{18,34} Patients with a score of ≤ 4 (i.e., seizure-free or nondisabling seizures only) were considered to have an excellent outcome. We have previously validated this criterion for excellent outcome by demonstrating that it is associated with the best work outcome, with ability to drive, and with improvement in independent living activities after surgery.³⁵ Favorable outcome was defined as improvement in seizure frequency score by ≥ 2 (i.e., $>75\%$ reduction in seizure frequency).

The patients were divided into two groups for analysis of the relationship between SISCOM localization and surgical outcome. The first group consisted of patients in whom SISCOM localization was concordant with the site of

Table 1 Seizure frequency scoring system used for analyzing postoperative outcome with respect to seizures

Seizure frequency	Score*
Seizure-free, not taking antiepileptic drugs	0
Seizure-free, need for antiepileptic drugs unknown	1
Seizure-free, requires antiepileptic drugs to remain so	2
Nondisabling simple partial seizures	3
Nondisabling nocturnal seizures only	4
No. of disabling seizures	
1–3 per year	5
4–11 per year	6
1–3 per month	7
1–6 per week	8
1–3 per day	9
4–10 per day	10
>10 per day	11
Status epilepticus	12

Modified from Engel et al.¹⁸ with permission of Lippincott Williams & Wilkins.

* Patients with a score of ≤ 4 (i.e., seizure-free or nondisabling partial seizures only) were considered to have an excellent outcome, and those whose scores improved by ≥ 2 (i.e., $>75\%$ reduction in seizure frequency) were considered to have a favorable outcome.

surgical resection. The second group consisted of patients in whom SISCOM and the resection site were nonconcordant or the SISCOM images were nonlocalizing. Comparison between the two groups was made with regard to the following: postoperative seizure frequency score, the proportion with excellent outcome, the proportion with favorable outcome, and the change in the seizure frequency score. Logistic regression analysis was performed to determine the independent contributions of SISCOM, MRI, and ictal scalp EEG in predicting surgical outcome.

Subgroup analysis was also performed among patients in whom MRI did not show a potentially epileptogenic lesion (i.e., “nonlesional” patients) to determine if the SISCOM was of prognostic value in this most challenging group of epilepsy surgery patients.

Correlation of the site and extent of SISCOM focus excision with surgical outcome. The two-SD subtraction SPECT images were coregistered with the postoperative MRI of patients with localizing SISCOM findings (figure). These images were presented to the two blinded primary reviewers, who were asked to assess whether the cerebral cortex underlying the hyperperfusion or hypoperfusion SISCOM focus had been completely resected, partially resected, or not resected. If there was disagreement between the reviewers, the third reviewer was asked to make an assessment. Postoperative outcome was compared among the three groups according to the rate of excellent outcome, the rate of favorable outcome, the postoperative seizure frequency score, and the degree of postoperative improvement in this score.

Statistical methods. The Mann-Whitney *U* test (two-tailed) was used in testing for significant differences between two ordinal or continuous variables, and Fisher’s

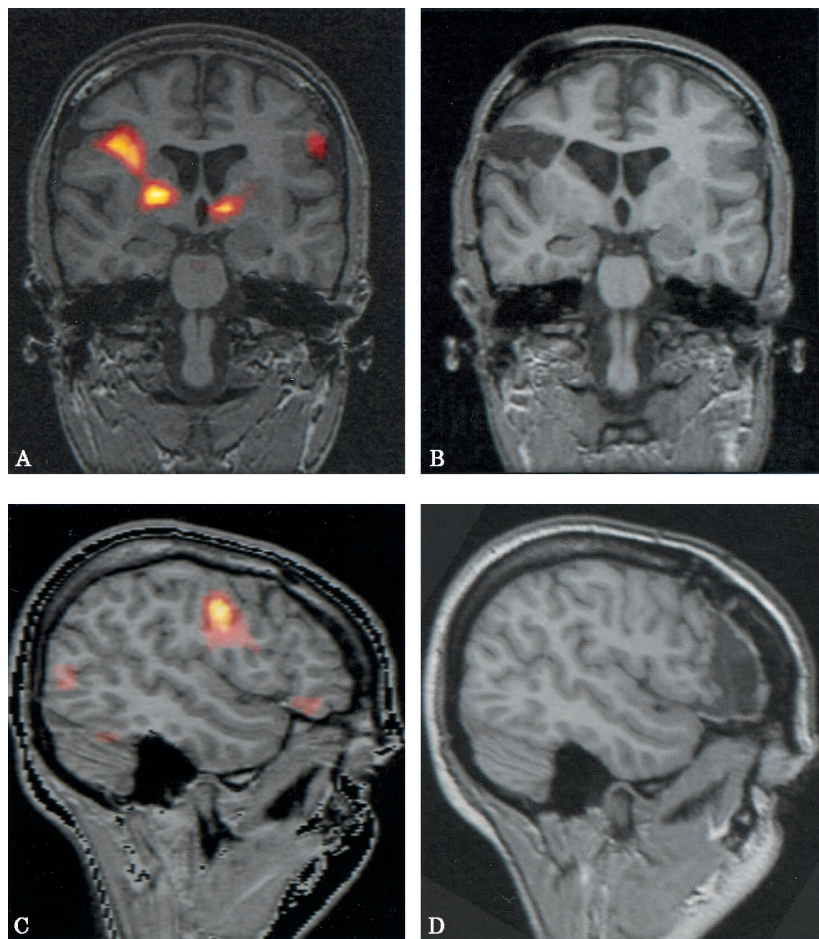


Figure. Preoperative subtraction ictal SPECT coregistered with MRI (SISCOM) images (A and C) and postoperative coregistered MRI (B and D). A and B, Complete excision of neocortical region underlying SISCOM focus. C and D, Non-excision of neocortical region underlying SISCOM focus.

exact test was used for dichotomous variables. Interobserver agreement between the primary reviewers was determined using Cohen kappa (κ) scores calculated for 19 possible choices.³⁶ Agreement was considered poor for $\kappa < 0.4$, good for $\kappa > 0.4$ but < 0.75 , and excellent for $\kappa > 0.75$.³⁷

For the analysis of the extent of the SISCOM focus resection, the χ^2 test was used for analysis of dichotomous variables (i.e., rates of excellent and favorable outcomes), and the Kruskal-Wallis ANOVA test was used for ordinal variables (i.e., postoperative seizure frequency score and improvement in this score).

Two groups of logistic regression analyses were performed using a maximal likelihood loss function. The first group used the SISCOM regional localization (concordant versus nonconcordant/nonlocalizing), preoperative MRI findings (lesional versus nonlesional), and ictal scalp EEG findings (localizing versus nonlocalizing) as the three independent variables. The second group of analyses used the extent of excision of the SISCOM focus, preoperative MRI findings, and ictal scalp EEG findings as the three independent variables. The dependent variables used for both groups of analyses were the achievement of excellent postoperative outcome and the achievement of favorable postoperative outcome.

Results. *Seizure and injection details.* The median length of the seizures was 72.5 seconds (range, 9 to 338 seconds). The median time from seizure onset to injection was 43 seconds (range, 6 to 182 seconds) and from injection

to the end of the seizure, 22 seconds (range, 0 to 164 seconds). Of the injections, 28 (78%) were ictal and 8 (22%) were postictal.

Blinded SISCOM review for cerebral localization. The interobserver agreement for SISCOM localization between the two primary reviewers is presented in table 2. Good interobserver agreement was found ($\kappa = 0.62$).

Comparison of SISCOM localization with EEG and MRI localizations. SISCOM, ictal scalp and intracranial EEG, and MRI results of the 36 patients are given in table 3. The scalp ictal EEG was localizing in 22 of 36 patients (61.1%), MRI in 19 of 36 (52.8%), and intracranial EEG recordings in 18 of 21 (85.7%). Of the 17 patients with localizing scalp EEG and SISCOM studies, the localiza-

Table 2 Comparison of interobserver agreement of seizure localization in 36 patients by two blinded reviewers for subtraction ictal SPECT coregistered with MRI

Localization	Agreed*	Disagreed
Localizing	18	11
Nonlocalizing	4	3
Total	22 (61.1%)	14 (38.9%)
$\kappa^\dagger$	0.62	—

* Both blinded reviewers localized the images to the same site or both agreed they were nonlocalizing.

† Cohen kappa score (κ) calculated for 19 possible choices (i.e., 18 possible brain regions plus nonlocalizing).

Table 3 Summary of EEG, MRI, and subtraction ictal SPECT coregistered with MRI (SISCOM) results and postoperative outcome results for 36 patients undergoing extratemporal epilepsy surgery

Patient	SISCOM localization	Scalp ictal EEG	MRI	Intracranial EEG	SISCOM focus excision	Surgical outcome*
1	R FT	R FT	UL	R FT	ICo	EX
2	L F	L F	UL	—	Co	EX
3	R F	R F	UL	R F	Co	EX
4	L F	L FT	L F (EM)	—	ICo	EX
5	R F	R F	UL	R F	Co	EX
6	R F	R FT	UL	R F	M	Non-Fa
7	L F	L F	UL	L F	ICo	Fa
8	R F	UL	UL	R F	M	Fa
9	L F	L F	L F (FCD)	—	ICo	Non-Fa
10	UL	R F	UL	R F	ICo	Fa
11	L F	UL	UL	L F	ICo	EX
12	UL	R F	R F (tumor)	—	UL	Fa
13	UL	UL	UL	R F	UL	Non-Fa
14	UL	R F	L F (tumor)	—	UL	Fa
15	UL	UL	F (EM)	R FP	UL	Non-Fa
16	R T	UL	UL	—	M†	Non-Fa
17	UL	L F	UL	—	UL	Non-Fa
18	L F	L C	L FP (tumor)	—	M	EX
19	UL	UL	R FP (tumor)	R F	UL	Fa
20	R F	R F	UL	R FP	ICo	EX
21	L FP	UL	L FP (EM)	—	Co	EX
22	UL	UL	R FP (FCD)	—	UL	EX
23	UL	UL	L FP (tumor)	UL	UL	Non-Fa
24	L F	L FT	L FT (EM)	—	ICo	EX
25	L T	R FT	R T (MTS)	R FT	M	Fa
26	L OP	UL	L OP (tumor)	—	M	Fa
27	UL	UL	R OP (EM)	—	UL	Fa
28	R TP	L OP	L OP (FCD)	—	M	EX
29	UL	UL	UL	UL	UL	Non-Fa
30	L OP	L PH	UL	L OP	ICo	Fa
31	R P	UL	UL	L & R OP	M	Non-Fa
32	L FP	UL	UL	L FP	ICo	Fa
33	R T	R PH	R TP (EM)	—	M	Non-Fa
34	R TO	R T	R TO (EM)	R TO	M	Fa
35	R T	R TP	R TP (tumor)	—	ICo	EX
36	UL	L PH	L TP (tumor)	L TP	UL	EX

* Excellent outcome (EX), postoperative seizure frequency score of ≤ 4 ; favorable outcome (Fa), improvement in score of ≥ 2 (i.e., 75% improvement in seizure frequency).

† Surgery performed at another institution.

R = right; FT = frontotemporal; UL = unlocalized; ICo = incomplete; L = left; F = frontal; Co = complete; EM = encephalomalacia; M = missed; FCD = focal cortical dysgenesis; T = temporal; C = central; FP = frontoparietal; MTS = mesial temporal sclerosis; OP = occipitoparietal; TP = temporoparietal; PH = posterior head; P = parietal; TO = temporo-occipital.

tions were concordant in 15 (88.2%). Of the 11 patients with localizing MRI and SISCOM studies, the localizations were concordant in 9 (81.8%). Of the 13 patients with localizing intracranial EEG and SISCOM studies, the localizations were concordant in 12 (92.3%).

SISCOM was localizing in 7 of the 14 (50%) patients in whom scalp ictal EEG was nonlocalizing. Of the 17 patients in whom MRI was nonlocalizing, SISCOM was localizing in 13 (76.5%). Of the 3 patients in whom intracranial EEG recording was nonlocalizing, SISCOM was localizing

Table 4 Comparison of postoperative outcome between patients with SISCOM localization concordant with surgical resection site and patients with nonconcordant or nonlocalizing SISCOM

Outcome	SISCOM localization		<i>p</i> Value
	Concordant, n = 19	Nonconcordant/nonlocalizing, n = 17	
Excellent, n (%)	11 (57.9)	3 (17.6)	0.019
Favorable improvement, n (%)	17 (89.5)	9 (52.9)	0.025
Postoperative SFS, median (range)	4 (0–9)	6 (1–10)	0.019
SFS improvement, median (range)	5 (0–10)	2 (0–7)	0.033

SISCOM = subtraction ictal SPECT coregistered with MRI;
SFS = seizure frequency score.

in 1 (33.3%) (in Patient 31, SISCOM was localizing to the right parietal region, but the subdural strip recordings suggested a bilateral occipitoparietal onset [table 3]).

Correlation of SISCOM localization with postsurgical outcome. The median postoperative follow-up was 16.5 months (range, 12 to 40 months). Overall, 14 patients (38.9%) had an excellent postoperative outcome and 26 (72.2%) had at least a favorable outcome. All patients with excellent outcome except one were seizure-free (aura not considered). The exception was a patient who had rare nocturnal seizures. The median preoperative seizure frequency score was 9 (range, 7 to 11), and the median postoperative score was 5 (range, 0 to 10). The median preoperative to postoperative reduction in this score was 3.5 (range, 0 to 10).

SISCOM localization was concordant with the site of the surgical excision in 19 patients (52.8%), nonconcordant in 5 (13.9%), and nonlocalizing in 12 (33.3%). The preoperative seizure frequency score was not different between patients in whom SISCOM localization was concordant

Table 5 Comparison of postoperative outcome in 17 patients with nonlocalizing MRI findings, according to presence or absence of concordance of SISCOM finding with surgical resection site

Outcome	SISCOM localization		<i>p</i> Value
	Concordant, n = 11	Nonconcordant/nonlocalizing, n = 6	
Excellent, n (%)	6 (54.5)	0 (0)	0.043
Favorable improvement, n (%)	10 (90.9)	1 (16.7)	0.01
Postoperative SFS, median (range)	4 (0–9)	8 (5–9)	0.02
SFS reduction,* median (range)	5 (0–8)	0 (0–8)	0.01

* Reduction in seizure frequency score (SFS) from preoperative to postoperative period.

SISCOM = subtraction ictal SPECT coregistered with MRI.

Table 6 Comparison of postoperative outcome in 14 patients with nonlocalizing ictal scalp EEG, according to the presence or absence of concordance of SISCOM finding with surgical resection site

Outcome	SISCOM localization		<i>p</i> Value
	Concordant, n = 5	Nonconcordant/nonlocalizing, n = 9	
Excellent, n (%)	2 (40)	1 (11.1)	0.50
Favorable improvement, n (%)	5 (100)	3 (33.3)	0.03
Postoperative SFS, median (range)	5 (1–6)	8 (2–9)	0.027
SFS reduction,* median (range)	5 (2–10)	1 (0–6)	0.023

* Reduction in seizure frequency score (SFS) from the preoperative to postoperative period.

SISCOM = subtraction ictal SPECT coregistered with MRI.

with the site of surgery and those with nonconcordant or nonlocalizing SISCOM; for both groups the median score was 9 and the range was 7 to 11 ($p = 0.55$). The postoperative outcome with respect to seizure control for the two groups is given in table 4. The concordant SISCOM group had higher rates of excellent and favorable outcome than the nonconcordant/nonlocalizing group ($p < 0.05$) and also had a greater reduction in the preoperative to postoperative seizure frequency score ($p < 0.05$).

Among the 17 patients with nonlocalizing MRI findings (table 5) and the 14 with nonlocalizing scalp EEG (table 6), concordance of SISCOM findings with the site of surgical resection was also associated with better postoperative outcome than for those with nonconcordant/nonlocalizing SISCOM. The only difference between the two groups that was not statistically significant was the rate of excellent outcome in patients with nonlocalizing ictal scalp EEG findings (see table 6).

The results of the logistic regression analyses are summarized in table 7. Concordance of SISCOM localization with the surgical site was predictive of excellent postoperative outcome ($p < 0.05$), but neither MRI nor scalp ictal EEG findings were.

Correlation of the extent of SISCOM focus excision with postsurgical outcome. The results of the blinded reviews of the two-SD SISCOM images coregistered with the patients' postoperative MRI for determining the extent of excision of the cerebral cortex underlying the SISCOM focus are given in table 8. The two primary reviewers were in agreement for 22 of 24 patients (91.7%).

The results of the comparison of postoperative outcomes according to the extent of SISCOM focus excision are summarized in table 9. There was a significant difference in the rate of excellent outcome among complete excision, incomplete excision, and nonexcision. The rate of favorable outcome, postoperative seizure frequency, and degree of seizure reduction were all significantly different among the three groups of patients.

The results of the logistic regression analyses for predicting postoperative outcome using the extent of SISCOM

Table 7 Results of logistic regression models predicting outcome after extratemporal epilepsy surgery in 36 patients

Modality	Excellent outcome ($\chi^2 = 8.93$, $p = 0.03$)			Favorable outcome ($\chi^2 = 9.98$, $p = 0.02$)		
	Regression coefficient (SE)	OR	<i>p</i>	Regression coefficient (SE)	OR	<i>p</i>
Constant	−4.9 (2.3)	—	0.04	−8.1 (3.0)	—	0.01
SISCOM*	2.0 (0.9)	7.2	0.03	2.4 (1.1)	11.2	0.03
MRI†	0.9 (0.8)	2.3	0.32	1.5 (1.0)	4.5	0.14
Ictal scalp EEG‡	0.9 (0.9)	2.6	0.27	0.7 (0.9)	2.1	0.42

* Subtraction ictal SPECT coregistered to MRI (SISCOM) groups: localization concordant with surgical site vs nonconcordant localization or nonlocalizing result.

† MRI groups: focal structural lesion detected preoperatively vs no focal lesion.

‡ Ictal scalp EEG groups: localizing vs nonlocalizing.

focus excision, the MRI findings, and the ictal scalp EEG findings as the three independent variables are given in table 10. The extent of the SISCOM focus excision was predictive of both excellent and favorable outcomes ($p < 0.05$), but neither the MRI nor the scalp ictal EEG findings were.

Discussion. A major reason that ETLE surgery has a less satisfactory outcome than TLE surgery is the difficulty in identifying the surgical epileptogenic focus in extratemporal regions, especially when a structural lesion is not detected by MRI.^{19,38} Ictal scalp EEG recordings are often poorly localizing in ETLE, and many of these patients require intracranial EEG recording before epilepsy surgery can be performed. However, intracranial electrode implantation carries risks of serious morbidity that have been estimated to be 4% and higher, and fatalities have occurred rarely.^{27,39,40} Extensive coverage of all possible sites of seizure onset increases the risk of complications of intracranial electrode implantation. A noninvasive method for localizing the potential epileptogenic zone in nonlesional ETLE patients could limit the extent and improve the accuracy of implanting intracranial electrodes for delineating the EEG abnormality. Furthermore, such a noninvasive method offers the potential of limiting the surgical resection to an extent necessary for successful postoperative outcome, possibly without the need for intracranial electrode implantation and recording.

FDG-PET is a potential noninvasive preoperative test that could be used to define the extent of the surgical focus. Although FDG-PET has been shown to be highly sensitive and specific for lateralizing the

origin of intractable TLE,^{6,41,42} it appears to be much less sensitive in intractable ETLE, and many series have reported a localization rate as low as 30%.³ Other forms of PET imaging, such as imaging of neuroreceptors (particularly benzodiazepine receptors), may prove useful in this group of patients. However, neuroreceptor imaging with PET is still essentially a research procedure, and no series involving a large number of ETLE patients has been reported. Another potential noninvasive technique for localizing the surgical focus is peri-ictal SPECT, but few studies have reported its potential usefulness in intractable ETLE. Compared with TLE, ETLE has significantly more practical and theoretical difficulties for conducting ictal SPECT studies. Extratemporal seizures are often brief, hypermotor, and nocturnal, making ictal injection more difficult. The ictal EEG discharge tends to spread more quickly,³⁸ which results in a less focal and sustained increase in regional cerebral blood flow. There is also evidence from earlier studies that the “postictal blood flow switch” may occur earlier in ETLE than in TLE.^{2,11,14,43,44} Nonetheless, the introduction of stabilized radioisotopes, particularly ^{99m}Tc-bicisate, has allowed injections to be made significantly earlier in the ictal period, which may help overcome some of the limitations mentioned above.²⁹

Our results demonstrate that peri-ictal SPECT, analyzed with the aid of the SISCOM method, shows a localized abnormality in two-thirds of the patients with intractable ETLE. It is particularly encouraging that this localization rate was at least as high for the 17 patients in whom MRI did not show a structural lesion. A previous study⁴⁵ of patients with TLE suggested that the results of ictal SPECT may merely reflect the ictal discharge on the scalp EEG. In contrast, we found that one-half of the patients whose ictal scalp EEG was nonlocalizing had localized SISCOM abnormalities. Moreover, we found that the concordance of the SISCOM focus with the site of surgery is predictive of an excellent postoperative outcome, whereas neither the scalp EEG nor the MRI findings were. These results strongly argue that SISCOM provides localizing information that is in addition to and not redundant for that provided by

Table 8 Results of blinded review for extent of excision of cerebral cortex underlying SISCOM focus in 24 patients

Reviewer	Complete excision	Incomplete excision	Not excised
Reviewer 1	5 (20.8)	9 (37.5)	10 (41.7)
Reviewer 2	4 (16.7)	11 (45.8)	9 (37.5)
Final determination	4 (16.7)	10 (41.7)	10 (41.7)

Values are n (%).

SISCOM = subtraction ictal SPECT coregistered to MRI.

Table 9 Comparison of postoperative outcome by extent of excision of cerebral cortex underlying SISCOM focus in 36 patients

Excision	Excellent outcome, n (%)	Favorable improvement, n (%)	Postoperative SFS, median (range)	SFS reduction,* median (range)
Complete, n = 4	4 (100)†	4 (100)†	1 (0–4)‡	7.5 (3–10)§
Incomplete, n = 10	6 (60)†	9 (90)†	3.5 (1–9)‡	6 (0–8)§
Not excised, n = 10	2 (20)†	6 (60)†	6.5 (0–10)‡	2 (0–6)§
Nonlocalizing, n = 12	2 (16.7)	7 (58.3)	6 (2–9)	2.5 (0–7)

* Reduction in seizure frequency score (SFS) from preoperative to postoperative period.

† $p = 0.018$, χ^2 .

‡ $p = 0.019$, Kruskal-Wallis ANOVA.

§ $p = 0.014$, Kruskal-Wallis ANOVA.

SISCOM = subtraction ictal SPECT coregistered to MRI.

the more standard tests. New techniques must be proved to provide additional information before they can be used routinely as a preoperative evaluation procedure.

The ultimate standard of proof of correct localization of intractable partial epilepsy is the demonstration that surgical resection of the region improves seizure control. We previously demonstrated this in mixed groups of patients who had TLE or ETLE.^{11,12} In the current study, the importance of SISCOM localization in determining surgical outcome was confirmed in patients with intractable ETLE. We found that the concordance of the SISCOM focus with the site of surgery is associated with a significantly higher rate of excellent postoperative seizure control than for a nonconcordant or nonlocalizing SISCOM study. This finding also applies to patients with nonlocalizing MRI findings and patients with nonlocalizing ictal scalp EEG results. Further analysis of our data reveals that the extent of excision of the SISCOM focus is predictive of the success of ETLE surgery. Concordance of the SISCOM focus with the surgical site and the extent of the excision of the SISCOM focus are the only two predictors of excellent outcome after ETLE surgery. Many of the patients in our study had SPECT studies because their seizures were not well-localized with more standard tests. Thus, we have demonstrated that in such patients, SISCOM frequently localizes extratemporal seizures when neither EEG nor MRI can localize

them. The information provided by SISCOM in these patients is not redundant to that provided by other tests. However, SISCOM may not be as valuable in patients whose seizures are already adequately localized by EEG or MRI.

It should be noted that not only did all four patients who had complete excision of the SISCOM focus have an excellent outcome, but so did 60% of those with a partial excision. This indicates that in many cases the SISCOM focus may overestimate the extent of the surgical epileptogenic zone, and it may not be necessary to excise all the focus involved by the SISCOM abnormality. Another note of caution is the potential effect of seizure spread. Two of the patients in whom the SISCOM focus was not included in the surgical resection became seizure-free. In one of these patients (Patient 28 in table 3), the seizure commenced in a left occipital area of focal cortical dysgenesis but rapidly propagated to the contralateral temporal lobe, where it remained localized for almost 4 minutes before generalizing. The injection of radioisotope for the SPECT study occurred late during the ictal period (182 seconds), well after the seizure had propagated to the right temporal lobe. This resulted in false lateralization by SISCOM. In the other patient (Patient 18 in table 3), the SISCOM focus was correctly localized to the left frontal region, but despite the short injection time of 20 seconds, the focus was not located within the structural MRI abnormality. Nonetheless, resection

Table 10 Results of logistic regression models predicting outcome after extratemporal epilepsy surgery in 24 patients

Variable	Excellent outcome ($\chi^2 = 11.8$, $p = 0.008$)			Favorable outcome ($\chi^2 = 9.6$, $p = 0.14$)		
	Regression coefficient (SE)	OR	p	Regression coefficient (SE)	OR	p
Constant	−9.3 (4.2)	—	0.04	−8.8 (4.5)	—	0.06
SISCOM excision*	2.7 (1.2)	201.0	0.03	2.4 (1.4)	117.3	0.08
MRI†	1.5 (1.4)	4.7	0.28	1.3 (1.4)	3.7	0.37
Ictal scalp EEG‡	0.75 (1.4)	2.1	0.60	−0.45 (1.5)	0.1	0.76

* Subtraction ictal SPECT coregistered with MRI (SISCOM) excision groups: complete excision of cerebral cortex underlying SISCOM focus vs incomplete excision or nonexcision.

† MRI groups: focal structural lesion detected preoperatively vs no focal structural lesion.

‡ Ictal scalp EEG groups: localizing vs nonlocalizing.

of this abnormality resulted in complete seizure control. Other investigators have also cautioned that rapid seizure spread may occasionally result in falsely localizing ictal SPECT images.⁴⁶

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MRI, ¹H-MRS, and functional MRI during and after prolonged nonconvulsive seizure activity

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Article abstract—*Background:* Various structural and functional changes, such as focal edema, blood flow, and metabolism, occur in the cerebral cortex after focal status epilepticus. These changes can be assessed noninvasively by means of MRI techniques, such as fluid-attenuated inversion recovery (FLAIR), EEG-triggered functional MRI (EEG-fMRI), and proton MR spectroscopy (MRS). *Methods:* The authors report on a 40-year-old patient with nonlesional partial epilepsy in the left posterior quadrant in whom these MRI techniques were applied in an active seizure focus and repeated during a follow-up of 1 year. *Results:* FLAIR imaging taken at the time of status epilepticus showed a signal hyperintensity in the occipital region. ¹H-MRS of this cortical region showed elevated lactate, decreased *N*-acetylaspartate (NAA), and elevated choline (Cho). In the same region, EEG-fMRI revealed an area of signal enhancement. After seizure control, recovery of lactate and Cho was observed, whereas the NAA level remained reduced. The structural abnormality demonstrated on FLAIR disappeared within 3 months. *Conclusions:* Repetitive MRI with sensitive sequences during clinically critical periods may disclose the structural correlate in a previously nonlesional epilepsy case. Corresponding to the clinical evolution, reversible and irreversible focally abnormal metabolism can be determined with ¹H-MRS, reflecting both increased neuronal activity and neuronal damage.

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During the last decade, new MRI methods were developed that are becoming available to improve the noninvasive localization of epileptogenic focus. These recent MRI techniques include the fluid attenuated inversion recovery sequence (FLAIR),¹ which improves the sensitivity of detecting structural changes associated with hippocampal sclerosis as well as in remote subcortical and cortical regions in epilepsy.^{2,4}

Another new approach to localize the epileptic focus is the use of functional MRI (fMRI). Blood oxygen level dependent (BOLD) fMRI is able to detect blood flow changes related to neuronal activity and has been applied in epilepsy using the hemodynamic response concomitant to neuronal firing during seizure.^{5–7} The aim is to record the EEG while the patient is in the magnet in order to trigger the fMRI acquisition with epileptic activity.⁸ This method provides images of interictal dysfunction with a high spatial resolution, allowing a direct comparison with structural MRI.

Proton MR spectroscopy (¹H-MRS) has been an established tool in clinical epilepsy studies for almost a decade. It has been used mainly to detect and quantify *N*-acetylaspartate (NAA), which is considered a marker for neuronal integrity. A decrease of the NAA level indicates neuronal loss or dysfunction and has been observed in the hippocampus by many groups.^{9–11} Other metabolic changes such as an increase of choline-containing compounds (Cho) and creatine + phosphocreatine (Cr) in the hippocampus have been reported and were attributed to reactive astrocytosis.¹⁰ In the neocortex involving epileptic activity, metabolite alterations including elevated glutamine/glutamate signal (glx) and lactate have been observed as well.^{12–14}

Few studies have used the outlined techniques in combination to identify the epileptic focus in patient with the initial diagnosis of nonlesional extratemporal epilepsy. In fact, this patient group is the most difficult to treat surgically if this line of treatment

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